

Homework No. 3: Manual vs. Automated Process Discovery.

You are a consultant hired to assist EDHospital, a medium size hospital, to analyze their Emergency Room. They provided a process description and an event log that came from their information system (see eClass).

General Instructions

- PM4PY help docs: <https://pm4py.fit.fraunhofer.de/docs>
- You are to submit: a PDF file with the BPMN model (Part I), and a PDF report for Part II (based on PM4PY). The slides are to be edited and turned into an effective presentation to the head of the Emergency Room conveying your main findings (you do not need to present the slides in class). You are to contrast your model in Part I with the discovered models in Part II.
- Deadline: November 19, 12:00AM.

Part I - Conceptual Modeling. (5pts.) Model the following process using the BPMN modeling tool in Signavio (we shall later contrast it with the models that you will discover).

EDHospital is a large hospital that treats patients from the entire Northern area of its country. Patients arrive to the Emergency Department (ED) independently or via an ambulance. All patients must go through registration prior to acceptance (some of the patients use the help of their relatives due to their condition). Once registered, patients are sorted by a nurse via triage. This includes the collection of patient's vital signs and medical history. Patients are then categorized into "walking", "lying" or "critical" types with the latter being in a severe condition. "Critical" patients get a fast-track treatment after which they are hospitalized.

After triage, the first two types of patients wait to be examined by a physician, who can either provide them with immediate treatment or send them to one or more of the following steps: imaging, lab external and consultation by a specialist physician (e.g. expert in Orthopedics). After gathering

the information from the above (or after waiting for the treatment to make effect) the treating physician makes a decision – the patient may either return to one of the following options: another round of tests/consultations, be admitted to the hospital, or get discharged home.

Part II - PM4PY. (5pts.)

- (1pts.) Load the ED event log into PM4PY and comment on the main activities and paths. What is the most common sequence of activities from start to end (aka variant)? What is the most common activity? You can use log exploration functions or process discovery.
- (1pts.) Comment on the time perspective by providing: average durations, shortest activities, longest activities, longest paths and average delays.
- (1pts.) Filter the process to observe the most infrequent variants. Are these the shorter or longer cases? Comment on your findings.
- (2pts.) Apply alpha (you can choose different variations of alpha), heuristic, and inductive miners (with default parameters for the heuristic and inductive miners). Provide the models that you get out of the algorithms (preferably in BPMN) and comment on the resulting models. Did you learn anything from running these algorithms on the data? Are the resulting models useful? Compare to your BPMN model.

Good luck!